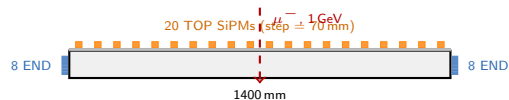


EndSparseTop: 15 ps timing with 20 sparse TOP SiPMs



Config	SiPMs	$\sigma(T_0)$
END+Vikuiti only	16	50.5 ps
EndSparseTop $N=20$	36	15.2 ps

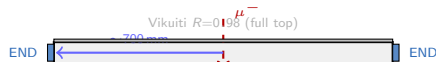
Estimator	Config	$\sigma(T_0)$
m^* -avg END (ref)	16 SiPMs	50.5 ps
TOP $k^*=3$	$N=20$	15.20 ps
BLUE (END+TOP)	$N=20$	15.21 ps

- **Factor 3.3× better** than END+Vik alone
- $w_{\text{TOP}} = 0.987 \Rightarrow \text{BLUE} \approx \text{TOP alone}$
- EJ-230 and EJ-204: **statistically equivalent**
diff = 0.07 ± 0.38 ps (0.2σ)
- Position range: 14.6–17.8 ps over ± 600 mm

All numbers verified independently from ROOT files.

What changed? Near-field vs far-field photon collection

END+Vikuiti (previous best):

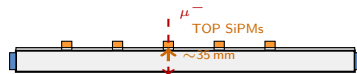


Photons travel ~ 700 mm before detection.

Effect	Impact
Long propagation	large time spread
Multiple reflections	Vikuiti losses
$\tau_{\text{prop}} \sim 3.7$ ns	propagation-limited

$\Rightarrow \sigma_{\text{END}} = 50.5$ ps (best with $m^* = 7$)

EndSparseTop (new):



TOP SiPMs capture photons after only ~ 35 mm.

	END	TOP ($N=20$)
Path	~ 700 mm	~ 35 mm
τ_{prop}	~ 3.7 ns	~ 185 ps
Limit	propagation	scintill. τ
σ measured	53.7 ps	15.2 ps

$\Rightarrow$ Near-field: short, uniform photon paths.

How the estimators are built

END: m -average

Sort photon times on left (L) and right (R) ends. Take mean of first m times on each side:

$$T_0^{\text{END}}(m) = \frac{\overline{t_L^{1\dots m}} + \overline{t_R^{1\dots m}}}{2}$$

Averaging reduces noise; too large m mixes in late photons. Optimal m^* .

Verified: $m^* = 8$, $\sigma_{\text{END}} = 53.68 \pm 0.93$ ps

TOP: k -th order statistic

Pool all arrival times on top face (all N SiPMs). Take the k -th earliest:

$$T_0^{\text{TOP}}(k) = t_{\text{TOP}}^{(k)}$$

$k=1$: maximum fluctuation from isolated early photons. $k=2-3$: plateau — noise reduced. Large k : longer paths included.

Verified: $k^* = 3$, $\sigma_{\text{TOP}} = 15.20 \pm 0.27$ ps

BLUE combination

Combine T_0^{END} and T_0^{TOP} with optimal weights:

$$T_0^{\text{BLUE}} = w_E T_0^{\text{END}} + w_T T_0^{\text{TOP}}$$

$$w_E = \frac{\sigma_T^2 - \text{cov}}{\sigma_E^2 + \sigma_T^2 - 2 \text{cov}}$$

Covariance measured event-by-event.

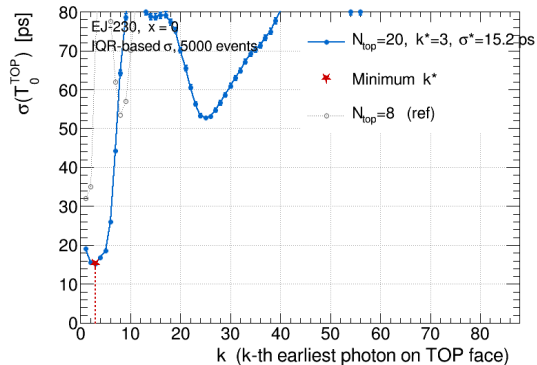
Quantity	Value
σ_{END}	53.7 ± 0.9 ps
σ_{TOP}	15.20 ± 0.27 ps
$\rho(T_{\text{END}}, T_{\text{TOP}})$	0.16
w_{END}	0.013
w_{TOP}	0.987
σ_{BLUE} (event-by-event)	15.21 ± 0.26 ps
σ_{BLUE} (from covariance)	15.18 ps ✓

Cross-check passed: $\sigma_{\text{BLUE}}^{\text{cov}} = 15.18$ vs $\sigma_{\text{BLUE}}^{\text{ev}} = 15.21$ ps.

$w_{\text{TOP}} \approx 1 \Rightarrow \text{BLUE} \approx \text{TOP alone.}$

Resolution: $\sigma_{\text{IQR}} = (Q_{75} - Q_{25})/1.349$.

Finding the optimal estimator



k -scan, TOP (EJ-230, $N=20, x=0$):

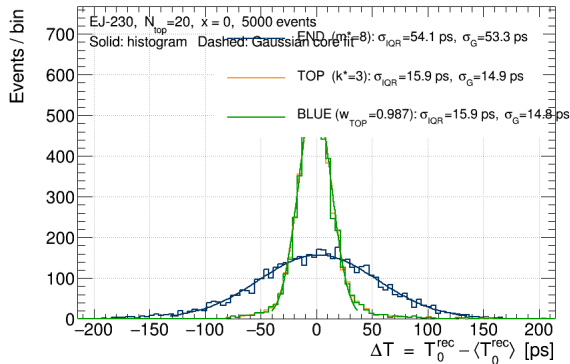
- $k=1$: earliest photon — large fluctuation
 - $k=2-3$: minimum — noise reduced
 - $k>5$: σ rises — longer paths enter
- $\Rightarrow k^* = 3, \sigma_{\text{TOP}} = 15.20 \pm 0.27$ ps

All N (EJ-230, $x=0$):

N	Total	m^*	k^*	σ_{TOP} [ps]
4	20	12	1	78.2 ± 1.8
8	24	9	1	32.0 ± 0.6
14	30	9	2	18.7 ± 0.3
20	36	8	3	15.20 ± 0.27

Higher $N \Rightarrow$ more near-field photons $\Rightarrow$ larger k^* .

The 15 ps result — residual distributions



EJ-230, $N=20$, $x=0$, 5000 events

	σ_{IQR}	σ_{Gauss}
END ($m^*=8$)	53.7 ps	53.3 ps
TOP ($k^*=3$)	15.20 ps	14.93 ps
BLUE	15.21 ps	14.80 ps

- END: Gaussian OK ($p=0.09$); $\sigma_{\text{IQR}} \approx \sigma_G$.
- TOP/BLUE: non-Gaussian tails; IQR is official metric.
- BLUE $\approx$ TOP: $w_{\text{TOP}}=0.987$; END negligible.

Timing comes almost entirely from TOP.

Why does it work? Near-field photon statistics

Photon yield (EJ-230, $x=0$):

Face	N_{pe}/ev	σ
LEFT end	514.9	~ 54 ps
RIGHT end	515.0	~ 54 ps
TOP ($N=20$)	1341	15.2 ps

Scale estimate (near-field limit):

$$\sigma \sim \frac{\tau_{\text{rise}}}{\sqrt{N_{\text{pe}}}} \xrightarrow{\text{EJ-230}} \frac{500 \text{ ps}}{\sqrt{1341}} = 13.7 \text{ ps}$$

Measured: 15.2 ps — consistent.

Sanity check only; k^* estimator uses order statistics, not a full average.

Nearest SiPM distance vs regime:

$n=1.58$, $v=c/n=190$ mm/ns

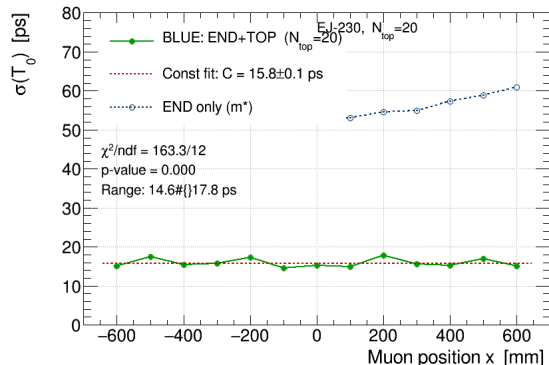
N	d_{max}	τ_{prop}	Regime
4	175 mm	920 ps	$\tau_p \gg \tau_r$
8	87.5 mm	460 ps	$\tau_p \approx \tau_r$
14	50 mm	263 ps	$\tau_p \sim \tau_r$
20	35 mm	184 ps	$\tau_p < \tau_r$

$\tau_r = \tau_{\text{rise}} = 500$ ps for EJ-230.

At $N=20$: propagation time $<$ rise time $\Rightarrow$ near-field regime.

For $N=4$ (nearest SiPM at 175 mm): propagation dominates and the $\tau/\sqrt{N_{\text{pe}}}$ model fails. $N \geq 14$ approaches the scintillation limit.

Position dependence — $N=20$, EJ-230



Fit and range (13 positions):

Min σ_{BLUE}	14.6 ± 0.3 ps ($x = -100$ mm)
Max σ_{BLUE}	17.8 ± 0.3 ps ($x = +200$ mm)
Peak-to-peak	3.2 ps (21%)

Const. fit C	15.75 ± 0.08 ps
χ^2/ndf	163/12
$p\text{-value}$	< 0.001

Interpretation:

- Variation **statistically real** ($\chi^2/\text{ndf} = 163/12$).
- Near SiPM (≤ 5 mm): $\sigma \approx 14.6\text{--}15$ ps;
at gap midpoint (~ 25 mm): $\sigma \approx 17\text{--}18$ ps.
- **3–4 $\times$ better than END+Vik** everywhere.
- Engineering: approx. uniform; NOT statistically flat.

BLUE weights: what does END actually contribute?

N	σ_{END} [ps]	σ_{TOP} [ps]	ρ	w_E	w_T	σ_{BLUE} [ps]
4	51.0	78.2	0.15	0.74	0.26	45.5 ± 0.8
8	50.2	32.0	0.18	0.21	0.79	28.9 ± 0.5
14	51.0	18.7	0.18	0.02	0.98	18.2 ± 0.3
20	53.7	15.2	0.16	0.013	0.987	15.2 ± 0.3

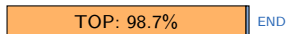
Correlation $\rho \approx 0.16$: low but non-zero. TOP SiPMs absorb some photons that would otherwise reach the END SiPMs — weak anticorrelation of yields.

BLUE cross-check ($N=20$):

$\sigma_{\text{BLUE}}^{\text{event}} = 15.21 \text{ ps}$ vs $\sigma_{\text{BLUE}}^{\text{cov}} = 15.18 \text{ ps}$ ✓

No bug. BLUE implementation correct.

Weight distribution at $N=20$:



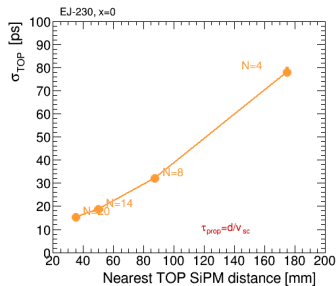
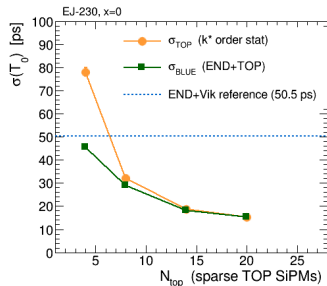
Key conclusion:

The improvement from 50 ps to 15 ps is driven **entirely by the TOP SiPMs**.

The 16 END SiPMs carry $< 1.3\%$ of the BLUE weight at $N_{\text{top}}=20$.

⇒ **Next step:** reduce END SiPMs (e.g. 4 per face instead of 8) while keeping $N_{\text{top}}=20$.
Expected σ penalty: small.

σ vs number of TOP SiPMs



Left panel:

σ_{TOP} and σ_{BLUE} vs N_{top} .
END+Vik ref (50.5 ps) dashed.

$\sigma_{\text{BLUE}} \approx \sigma_{\text{TOP}}$ for $N \geq 14$.

Right panel:

σ_{TOP} vs nearest SiPM distance.

$\tau_p = d/v_{\text{sc}}$ shown as dashed guide.

$N=4$ ($d=175$ mm): propagation dominates; $\tau/\sqrt{N_{\text{pe}}}$ model fails.

$N \geq 14$: $\tau_p < \tau_{\text{rise}}$, scintil. limit.

Diminishing returns:

$N: 8 \rightarrow 14$: -11 ps

$N: 14 \rightarrow 20$: -3.5 ps

EJ-230 vs EJ-204: are they really different?

Config	σ_{BLUE}	Uncertainty	$N_{\text{pe}}^{\text{TOP}}$
EJ-230, $N=20$	15.21 ps	± 0.26 ps	1341
EJ-204, $N=20$	15.14 ps	± 0.28 ps	1615
Difference	-0.07 ± 0.38 ps	(0.2σ)	

EJ-204 and EJ-230 are statistically equivalent at $N_{\text{top}}=20$.

No evidence for a real timing difference. EJ-204 has more top photons (1615 vs 1341), but the k^* -order-statistic already captures the minimum-time behaviour; extra photons give diminishing returns.

Why was EJ-230 recommended?

- 1 **Phase I:** EJ-230 better in END-only mode (50.5 vs 53.4 ps with m^* estimator).
- 2 **Speed:** $\tau=1.5$ ns vs 1.8 ns
— faster emission, benefits other regimes.
- 3 **Attenuation:** $L_{\text{att}}=120$ cm vs 160 cm
— less re-absorption at far end.
- 4 **Phase II:** No preference at $N=20$.
Both give 15.1–15.2 ps.

Recommendation:

Either scintillator is acceptable for EST $N=20$. Choose on cost, availability, non-timing criteria. Do **not** claim EJ-230 is timing-superior here.

Verified results (EJ-230, all from ROOT files):

END+Vikuiti reference	50.5 ps
EST $N=4$, BLUE	45.5 ± 0.8 ps
EST $N=8$, BLUE	28.9 ± 0.5 ps
EST $N=14$, BLUE	18.2 ± 0.3 ps
EST $N=20$, BLUE	15.21 ± 0.26 ps

Position uniformity ($N=20$):

Range 14.6–17.8 ps; 3.2 ps peak-to-peak.

Variation is real, correlates with SiPM spacing.

"Approximately uniform" — not statistically flat.

EJ-204 $\approx$ EJ-230: diff 0.07 ± 0.38 ps (0.2σ).

Four key messages:

- 1 **TOP SiPMs drive the improvement**
Near-field (~ 35 mm) vs far-field (~ 700 mm).
Factor $3.3\times$ better than END+Vik.
- 2 **EJ-204 and EJ-230 are equivalent**
at $N=20$. No timing preference.
- 3 **END SiPMs contribute $<1.3\%$ of BLUE weight**
at $N=20$. Timing is almost entirely determined by the TOP measurement.
- 4 **Next: reduce END SiPMs**
Explore 4 per face instead of 8.
Expected σ impact: negligible.